State of the art: speech recognition and speech databases, 2026

> **Draft for review, 2026-11-08.** The world in 2026 compared with 1995: what exists, what is free to use, and where open problems remain. The aim is to rebaseline Stage 2 before building anything, so that it does not reinvent what is already solved. Leaderboards change monthly; every figure here is dated, and any marked *check* still needs a primary source.

1. The short version

- **Transcribing speech is a solved commodity for major languages.** Open models with free weights transcribe English at about 5% average word error rate across mixed test sets. They identify the language, and cover from about 100 languages (Whisper) to more than 1,600 (Meta's Omnilingual ASR).
- **The 1995 task and the 2026 task are different problems.** The thesis classified one of eight known phrases. Today's systems transcribe any words. A 98% phrase-classification rate and a 5% word error rate cannot be compared directly.
- **Free data is plentiful.** LibriSpeech, Common Voice, FLEURS and Speech Commands are all free, and together cover read speech, crowd-sourced speech in more than 100 languages, and spoken digits.
- **The open problems are where this project's ideas already point:** accents, today's mix of telephone channels, short commands, learning without forgetting, and the same speaker decades apart. Section 5 maps each one.

2. Checking two claims

Two claims came from a conversation with Amazon's Alexa assistant.

Claim	Verdict
"Whisper was released as open source by OpenAI."	**Correct.** Code and weights are under the MIT licence (GitHub). It was released in September 2022; `large-v3-turbo` followed in October 2024.
"Whisper handles accent detection in the first 30 seconds."	**Partly wrong.** Whisper detects the **language**, not the accent, from the first 30-second window, and then transcribes the whole file in that language (detection guide, Phonexia comparison). It has no accent output. Strong accents can in fact make it pick the wrong language (section 5.1).
"LibriSpeech is available for free."	**Correct.** About 1,000 hours of read English from LibriVox audiobooks, CC BY 4.0, from OpenSLR SLR12 (Panayotov et al., ICASSP 2015).

3. Speech recognition models

Open models

The usual yardstick is the Hugging Face Open ASR Leaderboard. It averages word error rate (WER) over several English test sets. Its leader changed several times in 2026. Snapshots reported in May–November 2026 put the top models between about 4.8% and 5.5% average WER, with Whisper large-v3 near 7% (dev.to analysis, 2026-11-07, MarkTechPost, 2026-07). Read the live board before quoting a ranking.

Model	From	Languages	Licence	Why it matters here
Whisper large-v3, large-v3-turbo	OpenAI, 2022–2024	~100	MIT	The reference baseline everyone compares with. Built-in language ID.
Omnilingual ASR (300M–7B)	Meta, Nov 2025	1,600+, extendable by a few examples	Apache 2.0 (model), CC BY 4.0 (corpus)	The widest coverage; the closest thing to a "universal" recognizer (model card)
Cohere Transcribe, **Granite Speech 4.1**, **Qwen3-ASR**, **Parakeet / Canary**	Cohere, IBM, Alibaba, NVIDIA	mostly English plus a few	Cohere Transcribe Apache 2.0; others *check*	The 2026 accuracy leaders
wav2vec 2.0, HuBERT, WavLM	Meta, Microsoft	—	various	Self-supervised encoders: the feature extractors to compare formant ratios against (Phase 2)
VoxLingua107 models	—	107	CC BY 4.0 (data)	Language-ID baseline (Phase 3)

What changed since 1995

1995	2026
Hand-built features (formants, ratios)	Features learned from the raw waveform
One small network per speaker or vocabulary	One model for many languages and all speakers
Hundreds of recordings, made by the author	Hundreds of thousands of hours (Whisper's paper: 680,000)
Closed vocabulary, whole-utterance classification	Open vocabulary, word-by-word transcription
Minutes to train on a PC	Weeks on GPU clusters; but free to download and run

What 1995's approach still has that the giants lack: it learns from a handful of examples, it learns new classes **without forgetting old ones** (ART's stability–plasticity property), and it is small enough to explain. Those are the angles worth pursuing, not raw accuracy.

4. Speech databases

Free to use

Database	What	Size	Licence	Use in this project
LibriSpeech	Read English audiobooks	~1,000 h	CC BY 4.0	Standard English benchmark
Multilingual LibriSpeech	Read audiobooks, 8 languages	~50,000 h	CC BY (*check*)	Multilingual read speech
Common Voice	Crowd-sourced read sentences, with self-reported accent	137 languages (v22)	CC0	Languages and accents
FLEURS	The same sentences read in 102 languages	~12 h per language	CC BY	Language ID; like-for-like comparison across languages
People's Speech	Diverse English, commercial use allowed	~30,000 h	CC BY-SA / CC BY	Large English training set
Speech Commands	One-second words, **including the digits zero to nine**	~65,000 clips, 35 words (v2)	CC BY 4.0	Short-utterance and digit benchmark
Free Spoken Digit Dataset	Spoken digits 0–9	~2,000 clips, few speakers	CC BY-SA 4.0 (*check*)	Small digit sanity test
Speech Accent Archive	One paragraph read by speakers of 200+ native languages	~2,000+ speakers	CC BY-NC-SA (*check*)	Accent comparison on identical text
L2-ARCTIC	Non-native English: 24 speakers, 6 first languages	~27,000 utterances	CC BY-NC 4.0 (*check*)	Controlled accent identification
EdAcc	Conversations in English with diverse accents	40 h	*check*	Accented conversational speech
CommonAccent	Accent-labelled subset of Common Voice, with a SpeechBrain recipe	—	CC0 (from Common Voice)	Accent-ID baseline
VoxLingua107	YouTube speech in 107 languages	~6,600 h	CC BY 4.0	Language-ID training

Licensed (paid, through the LDC)

Database	What	Note
TIMIT	630 speakers, 8 US dialect regions	Contains phrase 2 of the thesis (sentence SA1)
TIDIGITS	Digit strings, 300+ speakers, ~13 h	The classic digit benchmark
Switchboard, Fisher, CallHome	Telephone conversations	Still the standard telephone benchmarks, but recorded 25–35 years ago on the old network

5. Where the open problems are

Each idea raised in this project, against what exists:

5.1 Accent identification (from the 1995 defence demonstration)

- **Exists:** accent classifiers (SpeechBrain's CommonAccent recipe) and labelled data (L2-ARCTIC, EdAcc, Common Voice, Speech Accent Archive).
- **Open:** a 2025 study found that **language-ID models are in effect accent classifiers** (arXiv 2506.00628). Heavily accented speech can be routed to the wrong language, so the wrong model loads. It is the 1995 demonstration's failure in a modern form: speaker independence did not mean accent independence then, and language ID does not mean accent robustness now.
- **Don't reinvent:** the classifier. **Do:** test the routing problem with the same phrases read in many accents.

5.2 Today's telephone channels

- **Exists:** reported effects of narrowband audio on modern models **disagree**, from almost no effect on Whisper large-v3-turbo (ru-asr-telephony) to 8–12 points of WER (Deepgram). Some benchmarks exist for real call audio (call-asr-bench). The standard telephone corpora are decades old.
- **Open:** no controlled, open comparison across **today's codec mix**: G.711, G.729, AMR-NB, AMR-WB, EVS and Opus, and real mobile and VoIP calls. That is the most clearly unclaimed gap in this survey, and the 1994 telephone database gives it a 30-year baseline.

5.3 Short utterances and spoken digits

- **Exists:** keyword spotting on Speech Commands is a mature benchmark, and digit recognition by modern models is routine.
- **Open for this project, not for the field:** making the formant-ratio front end work on one-word inputs. This is worth doing because the IVR goal (saying 0–9, * and # instead of pressing them) needs it, not because it would beat the state of the art.

5.4 Learning without forgetting

- **Exists:** large models are adapted with fine-tuning and adapters (LoRA), which can erase earlier skills ("catastrophic forgetting").
- **Open:** adding a speaker, accent or language **without retraining and without forgetting** remains an active research problem. Fuzzy ARTMAP's original selling point fits it directly.

5.5 The same voice 30 years apart

- **Exists:** speaker-ageing research uses mostly a few years' gap, or public figures' archival recordings.
- **Open:** a controlled same-speaker, same-phrase, same-microphone recording 32 years apart (1994 against 2026) is rare. The author's new recordings would make it a small but genuinely new data point.

5.6 Universal vocalization (Phase 4)

Already surveyed in STAGE2.md (NatureLM-audio, Project CETI, the Prat et al. fruit-bat dataset). Omnilingual ASR's few-example extension to new languages is the nearest human-speech analogue.

6. What this means for the plan

- **Use, don't build:**
  - Whisper and Omnilingual ASR as baselines;
  - LibriSpeech, Common Voice, FLEURS and Speech Commands as data;
  - the Open ASR Leaderboard's evaluation code for word error rates.
- **Build only where the gaps are:**
  - accent routing;
  - the modern telephone-channel comparison;
  - learning without forgetting;
  - the 30-year same-speaker recordings.
- **Report honestly:** the 1995 recognizer is a historical baseline. Its value now is in the questions it raises, not in competing on transcription accuracy.

Sources

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- Telephone channel: https://github.com/samishekhalard/ru-asr-telephony; https://github.com/leekery/call-asr-bench; https://deepgram.com/blog/noise-robust-speech-recognition-techniques; https://kili-technology.com/blog/asr-models-guide-word-error-rate-benchmarks-and-failure-modes-2026